Unit 5 historical timeline

public-student | All competition levels

Dates identify events, not an entire discovery compressed into one moment. Approximate ranges are intentional. Contextual contributors appear only where needed to explain the eight required scientists.

Date	Scientist	Event and significance	Source
c.1760–1762	Joseph Black	Development and presentation of specific/latent heat ideas Temperature alone does not measure thermal transfer.	H01,H02
1803	Black / editor John Robison	Posthumous collected lectures published Publication is later than the work and Black’s death.	H02
1824	Sadi Carnot	Book on motive power of heat Reversible-engine reasoning within a caloric framework.	H06
1834	Émile Clapeyron	Reformulation of Carnot’s work A documented route for transmission of ideas; not one of the eight required scientists.	H06
1840s	James Prescott Joule	Mechanical and thermal experiments Work and thermal effects are quantitatively related.	H05
1848	William Thomson / Kelvin	Absolute thermometric-scale proposal An early proposal, not the final modern Kelvin definition.	H03
1850	Rudolf Clausius	Mechanical theory of heat paper Reconciles engine reasoning with heat-work conversion.	H07
1860	James Clerk Maxwell	Gas distribution papers A population of molecular velocities replaces one common speed.	H10
1865	Rudolf Clausius	Entropy named and defined Developed over preceding years; distinct from energy.	H07

Date	Scientist	Event and significance	Source
1867 / 1871	James Clerk Maxwell	Demon privately described / publicly discussed Thought experiment probes microscopic control and boundaries.	H14
1877	Ludwig Boltzmann	Probability and entropy paper Statistical interpretation of thermal behavior.	H12
1900	Max Planck	Radiation theory and energy elements Beginning of quantum theory, not the completed photon picture.	H08,H09
1905	Albert Einstein	Light-quantum development Later contribution distinguished from Planck; contextual name.	H09
1918	Max Planck	Nobel Prize in Physics Recognition of earlier work, not the discovery date.	H08

Read the sequence critically

Black precedes the mature energy theory. Carnot precedes Joule’s experimental program. Clausius’s macroscopic entropy precedes Boltzmann’s statistical interpretation. Planck’s 1900 work precedes his 1918 award. Do not reorder these events to match a modern textbook’s chapter order. Source IDs resolve in U5-sources.md.